Health and the Economy: Analyzing the Effect of Income on Vaccination in US Counties in the Context of COVID-19

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I. Introduction

In this paper, we will assess the shocks associated with the recent COVID-19 epidemic by considering the effect of income on vaccination, controlling for education level, unemployment rates, political affiliation, and whether the observation region is metropolitan. In order to construct our analysis, we will use consider sample data for Covid-19 in the US at a county level, where data is recorded as of 11/15/2021.

The 2019-2020 Covid-19 pandemic exemplifies an unprecedent health shock that resulted in various responses and attempts to lessen the associated social and economic effects, namely, the release of the Covid-19 vaccine in December 2020. As we evaluate variation in vaccination rates across different counties, we consider potential barriers or social ideologies that might have made individuals in a population less able or less likely to get vaccinated. The effect of income on vaccination rates is particularly interesting considering that vaccinations were administered free of charge. While this might initially suggest that income should not be an explanatory variable of vaccination rates, there may still be characteristics or factors associated with different income groups that might make it possible to observe a relationship between income and vaccination in a county. For instance, a direct correlation between lower median income and lower vaccination rates might suggest implicit costs or accessibility barriers that are more likely to be experienced by lower-income households. Examples of such factors might include higher time constraints, cost concerns, transportation barriers, or other such accessibility constraints.

Identifying such a correlation may suggest a need to more closely consider underlying driving factors, particularly in the context of existing policies and systems that might magnify or lessen such relationship. Although our analysis focuses on Covid-19, a correlation between income and vaccination might have broader implications relating disparities in income and health equity in the US, which may be relevant when considering future economic policy. In addition, while Covid-19 corresponds to a specific instance in time, evaluating the factors that explain differences across our observations is useful in developing a more informed response that might help mitigate or prevent future shocks.

Table 1: Summary Statistics

Variable	Obs	Mean	Std. dev.	Min	Max
vax_number	3,115	57954.8	206114.7	0	6337170
vax_pct_18~s	3,115	54.12286	14.67371	0	99.9
Median_~2019	3,115	55624.28	14463.63	24732	151806
metro	3,115	.6266453	.4837728	0	1
adults_educ1	3,115	948058.3	4098148	450.2857	1.59e+08
adults_educ2	3,115	2124990	5707351	2267.605	1.57e+08
adults_educ3	3,115	2286280	6851281	1507.853	1.98e+08
adults_educ4	3,115	2547558	9175903	0	2.49e+08
voteshareD~6	3,115	.3155992	.1525608	.0314465	.9086382
Unemplo~2020	3,115	6.713965	2.241917	1.7	22.5

The number of vaccinated adults is the dependent variable of interest in this analysis. Although we will focus our regression analysis on the raw number of vaccinated individuals an observation unit, we may consider the number of vaccinations in the context of the total adult population for an observation (ie: vaccination rates, as shown in the second row of Table 1). Note that although the average vaccination rate in our sample data set is just above half of a county's population, the observed vaccination rates range from 0% to almost 100%.

The primary explanatory variable of interest is the median household income (reported for 2019), which has a sample mean of about $55,625. Again, we see a relatively large range across our sample data, from $24,732 to $151,806, which allows us to evaluate the impact of different levels of affluence in our observation units.

We also consider the sample statistics for other regressors. The row containing the metro dummy variable indicates that the probability that a county in our data set is metropolitan is about 62%. We also control for different education levels: *educ1* indicates adults with less than a high school degree, *educ2* indicates only a high school degree, *educ3* indicates some associates or college, and *educ3* indicates a bachelor's degree or higher. We can compare the numerical mean values in each education category, which suggest that on average, most of the population falls within *educ2* and *educ3*; nevertheless, we again see a broad range in the minimum and maximum values for each category. Additionally, we consider the impact of a county's political affiliation, which may be particularly relevant in cases where a population's votes hare is heavily skewed towards either Democratic or Republican candidates. Although there will be a small percentage of the vote share that goes towards neither Democratic or Republican, for simplicity, we will consider a population as either Democrat or non-Democrat (where non-Democrat will act as the base group in our regression analysis). From Table 1, we see that the average vote share for Democrats in the 2016 election (note that 2020 was reelection year) is about 32%, (so on average a county has about 58% non-Democrat votes). Again, we see a broad range in our observations, from counties with only about 3% Democratic vote share to counties with up to 90% Democratic vote shares. Lastly, we consider the effects of 2020 unemployment rates. We note that the mean unemployment rate across all counties was about 6.7%, which is higher than the natural rate of unemployment of about 4.4% (as reported by the Congressional Budget

Office, 2022). This is not unexpected however, since we intentionally evaluate the sample data in the context of the COVID-19 shock. We observe a relatively large range of unemployment rates in our sample data, from 1.7% to 22.5% (although most of the data points are within two standard deviations of the mean).

Primary regression of vaccination on median household income, controlling for education levels, voteshare, whether a county is metropolitan, and unemployment rates:

$$\begin{aligned} vax\_number = \beta_0 + \beta_1 median\_household\_income\_2019 + \beta_2 metro + \beta_3 adults\_educ2 + \beta_4 adults\_educ3 \\ + \beta_5 adults\_educ4 + \beta_6 unemployment\_rate\_2020 + \beta_6 voteshareD2016 + \mathrm{u} \end{aligned}$$

II. Regression Analysis

Coefficients and Standard Errors 1: Primary Regression

	vax_number_18plus
Median_Household_Income_2019	-0.111
	(0.084)
=1 if Metro, =0 if Non-Metro	9526.906
	(2074.567)
adults_educ2	0.013
	(0.002)
adults_educ3	0.002
	(0.002)
adults_educ4	0.011
	(0.001)
vote share for Democratic candidate for President in 2016	-4646.947
	(7610.600)
Unemployment_rate_2020	-306.863
	(314.873)
Intercept	-4371.854
	(5557.316)
Number of observations	3115

$$\begin{aligned} \widehat{vax\_number} = -4371.854 - .111 median\_household\_income\_2019 + 0526.906\ metro + .013\ adults\_educ2 \\ + .002\ adults\_educ3 + .011\ adults\_educ4 - 309.863\ unemployment\_rate\_2020 \\ - 4646.947\ voteshareD2016 \end{aligned}$$

From the regression above, we use OLS to estimate the partial effects of the explanatory variables on the number of vaccinated adults based on our sample data. The constant does not have a meaningful interpretation in this case, as it is not reasonable to consider a negative number of vaccinated adults (if all other explanatory variables were zero). While the negative coefficient on the median household income variable suggests that a $1 increase in median household income results in a .111 unit decrease in the number of vaccinated adults, this negative relationship is not statistically significant, with a p-value of 0.184. Likewise, the estimated coefficients for the partial effects of the unemployment rate and the vote share for a Democratic candidate in the previous election (both estimating an inverse relationship with the number of vaccinated individuals) are not statistically significant, with p-values of .330 and .542, respectively.

Nevertheless, all the coefficients of the variables measuring education for levels 2 and 4 reported p-values of .000, suggesting statistical significance even at the 1% level. Since all coefficients on the education categories are positive, this suggests that having more individuals with more a high school degree or greater may have some positive effect on the number of vaccinated individuals in that population. The coefficient of .014 on the second education category estimates that the effect of having a population with a greater number of adults with just a high school degree increases the number of vaccinated individuals by .013 more compared to the effect of increasing the number of adults with less than a high school degree by one. The effect of increasing the number of adults with a bachelor's degree or higher (level 2) is comparable to that of increasing the number of individuals with a high school degree.

The coefficient for the dummy variable indicating whether a city is considered metropolitan is also statistically significant at the 1% level in the above regression, with a p-value of .000. The estimated coefficient suggests that on average, a metropolitan city has a higher number of vaccinated individuals (9527 greater number of vaccinated individuals for metropolitan cities). Although this relationship does not seem unreasonable, as a more urban/populated area may experience more pressure for vaccination in order to curb the effects of COVID-19, it is important to note that the reported heteroskedastic-robust standard error of 2075 for this variable is relatively high, so the partial effect of this explanatory variable may not be quite as high as the estimated coefficient suggests.

Coefficients and Standard Errors 2: Regressing on logs of dependent and independent variables

	log_vaxNum
log_income	1.817
	(0.135)
=1 if Metro, =0 if Non-Metro	-0.868
	(0.059)
adults_educ2	0.000
	(0.000)
adults_educ3	-0.000
	(0.000)
adults_educ4	-0.000
	(0.000)
vote share for Democratic candidate for President in 2016	2.547
	(0.210)
Unemployment_rate_2020	0.134
	(0.013)
Intercept	-11.765
	(1.522)
Number of observations	3105

$$\log(\widehat{vax\_num}) = -11.765 + 1.817 log(med\_income) - .868 metro + (1.25 * 10^{-7}) adults\_educ2 - (2.30 * 10^{-9}) adults\_educ3 - (2.58 * 10^{-8}) adults\_educ4 + .134 unemployment\_rate\_2020 + 2.547 voteshareD2016$$

Let us now consider the same regression, but now consider the logarithmic transformation of the dependent variable (number of vaccinated adults) and the median household income regressor. This change may allow us to better characterize the percent changes in the number of vaccinated individuals as a result of changes in the regressors. Similarly, it may be more useful to consider the partial effect of a percent change in the median income rather than a dollar increase as in the previous regression.

Note that under the values that were not reported as statistically significant in the previous regression now suggest statistical significance up to the 1% level; under this logarithmic model, all p-values are .000 except for the coefficient for the level 4 education, which is statistically significant at the 10% level (p-value: .006) and the statistically-insignificant coefficient on the group of adults with level 3 education (associates or some college) (p-value: .939), which does not seem unreasonable given that we have separate categories for both high school degree and bachelor's and higher (level 3 could probably be linked with level 4 since it is not exceptionally distinct). Overall, we also see that the range in the reported standard error values is also significantly lower/closer to zero than in the previous regression.

Although closer to a positive value of vaccinated individuals, the intercept of -11.765 still does not have a meaningful interpretation in this model (see reasons stated for previous regression model). Note that under this new model, we observe a positive, statistically significant relationship between median household income; as the median household income of a county increases by 1%, we observe an average increase of 1.817% in the number of vaccinated adults in that population. Interestingly, under this regression model, we now observe a negative relationship between metropolitan cities and vaccinated adults; a metropolitan city suggests an 86.8% decrease in vaccinated adults compared to a non-metropolitan city. This relationship seems somewhat surprising; in future extensions of this analysis, it might be interesting to more specifically consider other characteristic of metropolitan vs non-metropolitan cities under this data set that could contribute to this relationship.

The coefficients on the number of adults in education groups 2 and 4 are statistically significant at the 1% level—level 2 (higher education than base group) indicates a positive effect on the percentage of vaccinated individuals in that population while level 4 indicates a lower effect on the percentage of vaccinated individuals in that population. Nevertheless, we observe that the magnitude of these coefficients is close to zero, suggesting that these positive differences in the effects of specific education level groups as the education level increases are somewhat negligible. (A future analysis might find it more useful to reduce the number of distinct education groups by combining levels 2-4).

As with the median household income, although previously statistically insignificant, we now observe a positive relationship between the 2020 unemployment rate and the percentage of vaccinated individuals reported up until 2021 (statistically significant at the 1% level). The coefficient estimate suggests that as a county's rate of unemployment increases by a percentage point, we observe a 13.4% increase in the number of vaccinated individuals in that county. This relationship seems reasonable, as higher vaccination may logically be partially explained by a reaction to higher unemployment rates brought about by the economic COVID-19 shocks in 2020.

Lastly, we consider the partial effect of the vote share for Democratic candidates relative to that of non-democratic candidates. We observe a positive coefficient on the Democratic vote share variable, suggesting that a one percentage point increase in the vote share in favor of the Democratic candidate in the most recent 2016 election increases the percentage of vaccinated adults in the population by approximately 257%, on average. Although the direction of this relationship is not very surprising given the context of the political implications surrounding beliefs relating to the efficacy of vaccination as a response to COVID-19, the degree of this effect is quite high. Note however, that because the vote share is measured as a percentage of the total population in a county, a one-point percentage increase may be a relatively large increase if considered in terms of number of individuals, which may partially explain the large magnitude.

Coefficients and Standard Errors 3: Interaction of Metro and Unemployment (under log model)

	log_vaxNum
log_income	1.829
	(0.090)
=1 if Metro, =0 if Non-Metro	-1.579
	(0.130)
adults_educ2	0.000
	(0.000)
adults_educ3	-0.000
	(0.000)
adults_educ4	-0.000
	(0.000)
vote share for Democratic candidate for President in 2016	2.560
	(0.143)
Unemployment_rate_2020	0.057
	(0.016)
metro_unemp	0.107
	(0.018)
Intercept	-11.387
	(1.015)
Number of observations	3105

$$\begin{aligned}\log(\widehat{vax\_num}) = &-11.387 + 1.829 log(med\_income) - 1.579 metro + (1.42 * 10^{-7}) adults\_educ2 \\ &- (8.41 * 10^{-9}) adults\_educ3 - (2.93 * 10^{-8}) adults\_educ4 \\ &+ .057 unemployment\_rate\_2020 + 2.560 voteshareD2016 + .107 metro\_unemp\end{aligned}$$

In this third regression, we add an interaction term between metro and the 2020 unemployment rate to further explore the relationship between how the effect of unemployment rates on vaccinated adults might depend on whether that county is metropolitan or not.

Under this modified regression model, all the statistical significance of all relationships at the 1% level from the previous regression still hold except for unemployment rate, which is no longer statistically significant as the p-value increases from.051. However, we additionally observe a statistically significant coefficient on the interaction term between metro and unemployment (p-value of .000). This estimated coefficient suggests that the positive effect of unemployment rates on vaccination in a county yields an additional increase (10.7%) in the percentage of vaccinated individuals for counties that are metropolitan. That is, the increase in vaccinated adults as a result of unemployment rates is statistically significant only for metropolitan counties.

Coefficients and Standard Errors 4: Dropping Education Variables

	log_vaxNum
log_income	2.060
	(0.118)
=1 if Metro, =0 if Non-Metro	-1.076
	(0.053)
vote share for Democratic candidate for President in 2016	3.080
	(0.176)
Unemployment_rate_2020	0.172
	(0.014)
Intercept	-14.514
	(1.341)
Number of observations	3105

$$\log(\widehat{vax\_num}) = -14.514 + 2.060log(med\_income) - 1.076metro + .172unemployment\_rate\_2020 + 3.080voteshareD2016$$

In the above regressions, although two of the three categories did show statistically significant differences from level 1 education, we observed very small values for the coefficients on all three education regressors. Thus, we now consider the regression model that drops those three variables.

Under this regression, all reported p-values are .000, suggesting that each explanatory variable still indicates a statistically significant partial explanation of vaccination in a county. Note that the standard errors do not vastly change, and the relationships still indicate the same direction in their relationship with the dependent variable. The most notable difference is that the magnitudes of the coefficients increase a bit, which is not surprising since our model now has less regressors to explain changes in vaccination (so the remaining explanatory variables will need to account for more of the observed changes). Nevertheless, these increases are relatively uniform and do not vastly vary from the regression that included the education regressors.

Let us test the exclusion restriction for the dropped variables on our unrestricted model against the alternative that the coefficients are not equal to zero:

$$H_0: \beta_{adults\_educ2} = \beta_{adults\_educ3} = \beta_{adults\_educ4} = 0$$

The F-statistic resulting from this test on these three exclusion restrictions is 196.94, with a p-value of .0000. Thus, at the 1% significance level, we reject the null hypothesis that the collective effect of these variables is not significant in explaining vaccinations.

Thus, we conclude that the education variables are jointly significant. We should not drop this group of variables from the model, as at least one of them has a partial effect on y. Note that even though we conclude that these variables are jointly significant, we observed individual insignificant relationship for some of these variables.

III. Variance and Homoskedasticity Assumption

Note that in the above regressions, we calculate the standard error values using heteroskedastic-robust estimates for calculations of standard errors. From the Breusch-Pagan test, we find that the homoskedasticity assumption does not hold in the models calculated above; that is, we do find statistically significant evidence suggesting the variance in our x values is not constant across our sample data set. Consider the Breusch Pagan test for the Primary Regression model as described below.

Let us maintain the assumption that the expected value of the error term is zero; then the variance of u is equal to the expected value of the squared residual error term.

Under this assumption, we can test the null hypothesis that the square of the residuals—that is, the variance of the error term—is constant across our samples. We test this claim against the alternative hypothesis that the variance in the error term can be expressed as a linear function of the explanatory variables. Mathematically, given:

$$\hat{u}^2 = \beta_0 + \beta_1 median\_household\_income\_2019 + \beta_2 metro + \beta_3 adults\_educ2 + \beta_4 adults\_educ3 + \beta_5 adults\_educ4 + \beta_6 unemployment\_rate\_2020\,) + \beta_7 voteshareD2016 + \mathrm{v}$$

$$H_0: \beta_0 = \beta_1 = \beta_2 = \beta_3 = \beta_4 = \beta_5 = \beta_6 = \beta_7 = 0$$

Running an F-test on the seven exclusionary restrictions for the model regressing the estimates for the squared residual error terms on the explanatory variables from the primary regression results in an F-statistic of 178.98, with a p-value of 0.0000. As the p-value 0.000 is significant at the 1% level, we reject the null hypothesis that the variance of the error term is constant across our observations; thus, we can estimate the standard errors in our model as usual.

We can run a similar process on our related regression models. Running the same model using the log form of the number of vaccinations and the median household income results likewise results in a p-statistic of 0, with an F-statistic of 64.39. The regression model interacting unemployment and metro had an F-statistic of 77.54, p-value of .000 under the Breusch-Pagan test, and the model that dropped variables had an F-statistic of 18.49 with a p-value of .000. Thus, we reject the assumption of homoskedasticity in all cases, and use heteroskedastic robust calculations for standard errors.

IV. Joint Hypothesis

Let the model estimated in the primary regression (Regression 1) be the unrestricted model of interest:

$$vax\_number = \beta_0 + \beta_1 median\_household\_income\_2019 + \beta_2 metro + \beta_3 adults\_educ2 + \beta_4 adults\_educ3 + \beta_5 adults\_educ4 + \beta_6 unemployment\_rate\_2020 + \beta_6 voteshareD2016 + \mathrm{u}$$

In the analysis of this model, we observed that none of the following explanatory variables had significant p-values: median household income, 2020 unemployment rate, and vote share for the Democratic candidate in the previous election. Let us now consider the joint hypothesis that, controlling for other regressors in our model, this set of variables has no effect on vaccination numbers. Consider the restricted model under this hypothesis:

$$vax\_number = \beta_0 + \beta_2 metro + \beta_3 adults\_educ2 + \beta_4 adults\_educ3 + \beta_5 adults\_educ4 + \text{u}$$

Coefficients and Standard Errors 5: Restricted Model

	vax_number_18plus
=1 if Metro, =0 if Non-Metro	11273.132
	(3283.592)
adults_educ2	0.013
	(0.002)
adults_educ3	0.002
	(0.003)
adults_educ4	0.011
	(0.001)
Intercept	-1.51e+04
	(3737.019)
Number of observations	3115

Compared to the unrestricted model, we see that the both the magnitude and the robust-standard error of the intercept and of the coefficient on metro and increase. The direction of all relationships between explanatory variables and vaccination does not change, and the coefficients and error terms on the education variables remain unchanged from the unrestricted model. However, the p-value of metro increases from .000 to .001 (still significant at the 1% level).

We test the joint hypothesis that the set of variables has no effect on vaccination:

$$H_0: \beta_1 = \beta_6 = \beta_7 = 0$$

The resulting F-statition for the three exclusion restrictions is 0.75, with a p-value of .5219. Thus, we fail to reject the null hypothesis that the set of variables have no effect. Based on this joint hypothesis test, we may consider dropping median household income, democratic vote share, and unemployment regressors from our model.

V. Conclusion

The primary regression model suggests that income does not have a significant effect on vaccination; we find evidence in our joint hypothesis test to drop this (as well as several other) regressors from the model that explains changes in the number of vaccinated adults in a population. Nevertheless, we find that although there is not strong evidence for a linear relationship between number of vaccinated adults and income, when we perform a log transformation on the number of vaccinations and the median household income, we find strong evidence that median household income has a positive effect on vaccination. Namely, the sample data set suggests that on average—controlling for a county's education levels, vote share, unemployment, and metropolitan status—a county with 1% higher household income will have approximately 1.8% greater number of vaccinated adults.

While the inclusion of other regressors helps reduce the unobserved partial effects when considering changes in vaccination rates across our observations, it is still important to note that there is likely a degree of randomness/unobservables when explaining why certain counties might have been more heavily impacted by Covid-19 (which in turn might contribute to vaccination rates). Nevertheless, as we evaluate patterns between income and vaccination at the 1% significance level, the analysis still suggests strong evidence indicating a positive correlation.